

Lab 13: Financial Risk System - Architecture Design

a. Architectural characteristics

Based on the requirements, the key architectural characteristics are:

1. Scalability:
 - Justification: Trade volume grows daily and must be supported for at least 5 years without redesign.
 - Priority: High
 - Requirement: The system must scale to process increasing trade volumes (initial ~5,000 trades, +10/day) within the nightly batch window.
2. Availability:
 - Justification: Business users across time zones depend on reports daily before trading begins.
 - Priority: High
 - Requirement: Risk reports must be accessible 24x7 with less than 30 minutes of downtime per day.
3. Performance: All processing must complete between 5pm New York and 9am Singapore.
 - Justification: Reports must be generated between 5pm New York and 9am Singapore.
 - Priority: High
 - Requirement: The system must complete import, enrichment, calculation, and report generation within the overnight processing window.
4. Security: Authenticated and authorized access only. Role-based access (read-only users vs parameter editors).
 - Justification: Financial risk data is highly sensitive and access is restricted by bank policy.
 - Priority: High
 - Requirement: All access must be authenticated and authorized, with role-based permissions for viewing reports and modifying parameters.
5. Auditability & Traceability: Must track report generation, parameter changes, and input data.

- Justification: The bank must explain how risk values were calculated for compliance and investigation.
 - Priority: High
 - Requirement: The system must record report generation, parameter changes, and retain input data for traceability.
6. Reliability & Fault Tolerance: System should recover from errors when possible and log all errors.
- Justification: Batch failures must not silently block business operations.
 - Priority: Medium
 - Requirement: Errors must be logged, partial recovery attempted, and critical failures reported via SNMP.
7. Maintainability & Extensibility: Reference Data System will be replaced in ~3 months.
- Justification: The Reference Data System will be replaced within 3 months.
 - Priority: Medium
 - Requirement: The architecture must allow reference data sources to be replaced with minimal impact.
8. Operability & Monitoring: SNMP alerts for fatal errors and missed report deadlines.
- Justification: Operations teams require visibility into failures and SLA breaches.
 - Priority: Medium
 - Requirement: SNMP traps must be sent for fatal errors or missed report deadlines.
9. Data Retention: Input files must be retained for 1 year.

b. Detailed scenarios (quality attribute scenarios)

Scenario 1 – Availability

Source: Business user in Singapore

Stimulus: User attempts to access risk report at 8:30am Singapore time

Environment: Normal operations, after batch processing completes

Artifact: Risk Report Web Portal

Response: System authenticates the user and displays the latest completed risk report

Measure: Report is displayed within acceptable response time and before 9am

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Scenario 2 – Performance

Source: Trade Data System

Stimulus: Large trade XML file arrives at 5pm New York

Environment: Nightly batch processing window

Artifact: Trade Import and Risk Calculation Engine

Response: System imports, enriches, calculates, and generates reports

Measure: All reports generated before 9am Singapore time

Scenario 3 – Security

Source: Unauthorized business user

Stimulus: User attempts to modify risk calculation parameters

Environment: Normal system operation

Artifact: Configuration Management UI

Response: System denies access and logs the attempt

Measure: No parameter change occurs and audit log entry is recorded

Scenario 4 – Auditability

Source: Compliance officer

Stimulus: Requests explanation of a counterparty's risk value

Environment: Post-report generation

Artifact: Audit Log and Data Retention Store

Response: System provides input files, parameters, and timestamps used

Measure: Full traceability without manual reconstruction

Scenario 5 – Reliability

Source: Reference Data Importer

Stimulus: Corrupted counterparty XML file

Environment: Nightly batch processing

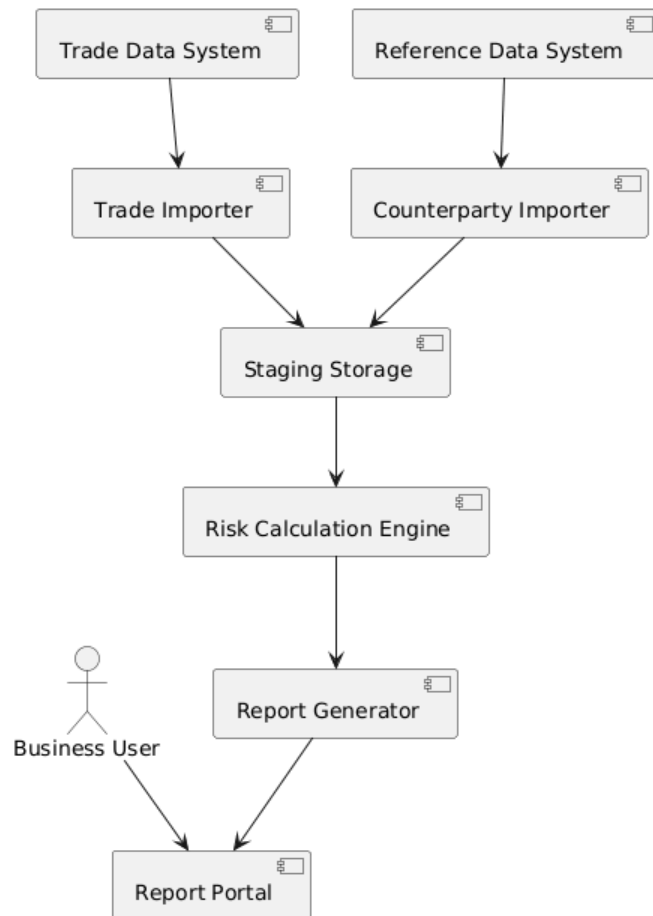
Artifact: Import Module

Response: System logs error, skips invalid records, continues processing valid data

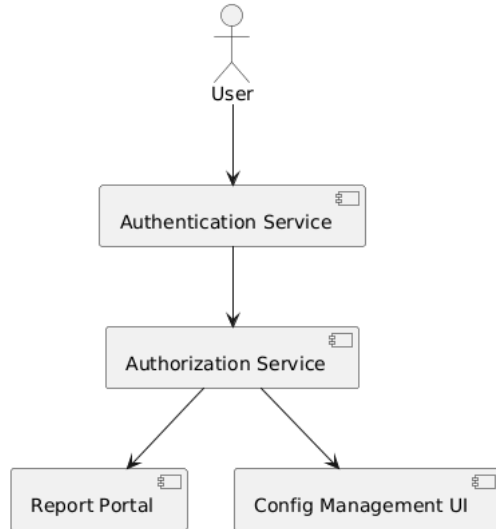
Measure: Report still generated with warning logged

c. Architecture diagrams (PlantUML)

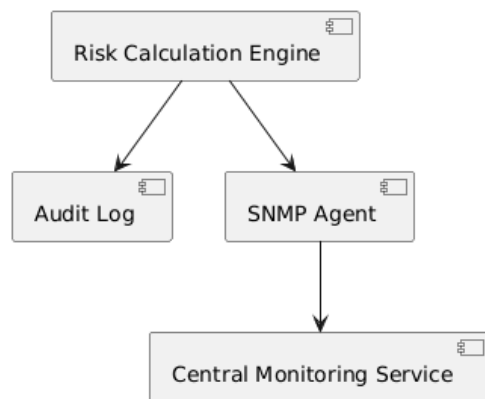
1. Overall System Architecture



2. Security & Authorization Architecture



3. Audit & Monitoring Architecture



Presentation Layer

- **Web Framework:** Spring Boot with Thymeleaf or React.js
- **Web Server:** Apache Tomcat or Nginx
- **Load Balancer:** HAProxy or AWS ELB

Application Layer

- **Application Server:** Spring Boot (embedded Tomcat)
- **Batch Processing:** Spring Batch
- **Scheduler:** Spring Scheduler or Quartz
- **Authentication:** Spring Security with LDAP integration

Integration Layer

- **XML Parsing:** JAXB or Jackson XML
- **SNMP Client:** SNMP4J library
- **File Monitoring:** Apache Commons VFS or Spring Integration

Data Layer

- **Primary Database:** PostgreSQL or Oracle Database
- **Caching:** Redis for session management
- **File Storage:** NFS or Amazon S3 for archives
- **ORM:** JPA/Hibernate

Cross-Cutting

- **Logging:** SLF4J with Logback
- **Monitoring:** Spring Boot Actuator + Prometheus + Grafana

d. Risks of the proposed architecture (detailed)

Risk 1: Batch processing may not complete before 9am Singapore deadline

Description:

Batch processing may not complete within the available processing window (5pm New York to 9am Singapore).

Impact:

High – Business users cannot access risk data before the trading day starts.

Probability:

Medium

Factors:

- Data quality issues causing reprocessing or retries
- Unexpected increase in trade volume
- System performance degradation over time
- Network latency during file transfers
- Increasing complexity of risk calculations

Consequences:

- Traders operate without current risk visibility
- Potential regulatory compliance issues
- Loss of business confidence in the system
- Manual workarounds required by operations teams

Risk 2: Single batch job failure blocks report generation

Description:

Failure in one batch component (import, enrichment, or calculation) may halt the entire report generation process.

Impact:

High – No risk reports are produced for all users.

Probability:

Medium

Factors:

- Tight coupling between batch steps
- Lack of automatic retry or recovery mechanisms
- Corrupted input files
- Insufficient error isolation

Consequences:

- Complete loss of daily risk report
 - Increased operational support workload
 - Delayed business decisions
 - Emergency manual calculations
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Risk 3: Incorrect or incomplete input data affects risk accuracy

Description:

Invalid, missing, or inconsistent trade or counterparty data may lead to incorrect risk calculations.

Impact:

High – Incorrect risk exposure may lead to poor trading decisions.

Probability:

Medium

Factors:

- File-based data transfer without real-time validation
- Inconsistent reference data
- Partial or corrupted XML files
- Late arrival of input files

Consequences:

- Incorrect exposure calculations
 - Regulatory audit findings
 - Financial losses
 - Reduced trust in reported figures
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Risk 4: Security misconfiguration exposes sensitive data

Description:

Incorrect access control configuration may allow unauthorized users to view or modify sensitive risk data.

Impact:

High – Confidential financial data exposure.

Probability:

Low to Medium

Factors:

- Misconfigured roles and permissions
- Human error during user provisioning
- Lack of periodic access reviews
- Overly broad authorization rules

Consequences:

- Data confidentiality breach
 - Regulatory penalties
 - Reputational damage
 - Mandatory security investigations
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Risk 5: Tight coupling to current Reference Data System

Description:

Strong dependency on the existing Reference Data System may complicate migration to the new system.

Impact:

Medium – Increased effort and risk during system transition.

Probability:

High

Factors:

- Hard-coded file formats or schemas
- Limited abstraction of reference data access
- Insufficient interface boundaries

Consequences:

- Delayed integration with new reference system
 - Increased development and testing effort
 - Risk of production outages during migration
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Risk 6: Insufficient monitoring and delayed failure detection

Description:

Failures may not be detected immediately, delaying corrective action.

Impact:

Medium – Problems escalate before being addressed.

Probability:

Medium

Factors:

- Limited real-time monitoring
- Dependence on manual checks
- Insufficient alert coverage

Consequences:

- Late awareness of missed deadlines
- Increased downtime
- SLA breaches
- Operational inefficiency

e. Risk mitigation strategies

Mitigation for Risk 1: Batch processing may not complete before 9am Singapore deadline

Strategy:

Improve batch performance, resiliency, and predictability.

Actions:

- Parallelize trade processing by counterparty or trade batches
- Optimize database indexing and calculation logic
- Introduce performance monitoring for each batch phase
- Pre-validate input files to fail fast on bad data
- Allow partial reprocessing instead of full reruns

Residual Risk:

Low to Medium – Dependent on unexpected spikes in data volume.

Mitigation for Risk 2: Single batch job failure blocks report generation

Strategy:

Increase fault tolerance and isolation between batch steps.

Actions:

- Split batch processing into independent, restartable stages
- Persist intermediate results after each stage
- Implement automatic retries with limits
- Allow last successful report to remain available if batch fails
- Add manual restart capability for failed steps

Residual Risk:

Low – Failures become localized and recoverable.

Mitigation for Risk 3: Incorrect or incomplete input data affects risk accuracy

Strategy:

Strengthen data validation and traceability.

Actions:

- Enforce XML schema validation on import
- Quarantine invalid records instead of aborting entire batch
- Log data quality warnings in audit logs
- Include data completeness indicators in reports
- Retain raw input files for investigation

Residual Risk:

Medium – External data sources remain a dependency.

Mitigation for Risk 4: Security misconfiguration exposes sensitive data

Strategy:

Apply strict access control and governance.

Actions:

- Enforce role-based access control (RBAC)
- Separate read-only users from parameter administrators
- Apply principle of least privilege

- Log all access attempts and configuration changes
- Perform periodic access reviews

Residual Risk:

Low – Human error still possible but detectable.

Mitigation for Risk 5: Tight coupling to current Reference Data System

Strategy:

Decouple reference data access through abstraction.

Actions:

- Introduce adapter or interface layer for reference data
- Isolate file parsing logic from business logic
- Support multiple reference data formats in parallel
- Test new reference system in shadow mode before cutover

Residual Risk:

Low – Migration risk reduced to adapter changes.

Mitigation for Risk 6: Insufficient monitoring and delayed failure detection

Strategy:

Improve observability and alerting.

Actions:

- Configure SNMP traps for batch failures and missed deadlines
- Add health checks for batch components
- Track batch execution times and trends
- Provide operational dashboards for overnight processing
- Alert operations teams before SLA breaches occur

Residual Risk:

Low – Issues detected early and acted upon.